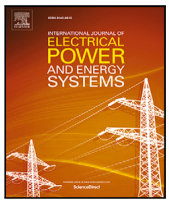
Electrical Power and Energy Systems 163 (2024) 110340



Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes





Peak interval-focused wind power forecast with dynamic ramp considerations

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ARTICLE INFO

Keywords:
Wind power forecast
Local peak interval
Wind power ramps
Deep learning

ABSTRACT

Wind power forecast (WPF) plays a crucial role in the reliable and safe operation of power systems. Existing wind power prediction methods mainly focus on the overall trend information, often neglecting the performance of some typical intervals that can significantly impact the reserve requirements of power system, ***e.g.***, unexpected ramp-up and ramp-down, local peak interval (LPI), and so on. Therefore, the primary focus of this paper is to explore techniques for enhancing the accuracy of wind power during the LPI. Considering that the points during the LPI are influenced by dynamic ramp characteristics, with the steepness and magnitude of ramps influencing the range of LPI, this paper proposes a novel focused-LPI model by incorporating dynamic ramp considerations. Specifically, beyond the tradition forecasting model, the proposed model encodes the information of ramp occurrence in wind power data based on a positional encoder; subsequently, the proposed model not only outputs standard predictions within a prediction range but also identifies the presence of ramps within these predictions. In addition, this paper employs a dynamically weighted loss function to optimize the proposed multi-task model. The proposed model is applied for various network architectures, and the results confirm that the proposed model is widely suitable for various basis network architectures. Specifically, the R-squared (R2) of the proposed model for the overall trend can exist a little error compared to the initial basic network architectures; the Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Relative Error (MRE) of the proposed model for predicting the LPI points can outperform the initial basis network architecture by average enhancements of 7.10%, 2.66%, and 3.46%, respectively. In addition, compared to existing state-of-the-art (SoTA) models in the LPI points, the proposed model obtains improved performance with 14.43% for MAE and 11.19% for RMSE, respectively, presenting its substantial performance enhancement.

1. Introduction

Forecasting wind power accurately is essential to ex-ante power system operation and control, e.g., economic dispatch and unit commitment. Recent global warming has led to increasingly complex meteorological conditions, accelerating the irregularities and uncertainties in wind power. This unpredictability in wind patterns results in multiple critical power intervals, including local peak interval (LPI), unexpected up and down ramps, and sudden fluctuations in wind power data. These critical intervals directly affect the stability and safety of the power system, making accurate forecasting essential for maintaining an efficient and reliable power grid. In particular, the points during the LPI significantly influences the specific reserve requirements of the grid. Therefore, it is rather essential to develop a advanced forecast model that focuses on the accuracy of wind power during the LPI points.

Currently, multi-step wind power predictions are typically calculated using deep learning-based methods to determine overall accuracy. Deep learning-based models are popular prediction tools due to their excellent capability of analyzing the nonlinearity and uncertainty of wind power features [1]. Generally, neural network-based deep learning-based models are decomposed into two components. One is the network architecture that plays a fundamental role in feature extraction, and another is the projection layer [2] that is responsible for implementing task-specific functionalities. The quantity and quality of the features extracted from the network architecture determine the performance of a specific task. Therefore, to capture more abundant features, the task of wind power forecast based on deep learning has evolved from the original simple single model to the recent more complex fusion models. Single models, including long short-term memory

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https://doi.org/10.1016/j.ijepes.2024.110340
Received 24 April 2024; Received in revised form 4 September 2024; Accepted 22 October 2024
Available online 4 November 2024
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(LSTM) [3], conventional Neural Network (CNN) [4], and attention [5], are widely adopted to predict wind power. Single models may have inferior performance in face of complex and uncertain data, failing to capture the intricate correlations and resulting in suboptimal performance [6]. Therefore, to extract deeper features from wind power data, researchers begin to investigate fusion models that combine multiple single models to improve the prediction accuracy. For example, Ref. [7] fused CNN and RNN to extract more features from different angles. Ref. [8] fused ResNet and LSTM to obtain superior performances than single models. Ref. [9] fused CNN and attention for wind speed forecasting. The results demonstrate enhanced accuracy than other models. Transformer is a powerful neural network architectures in capturing long-term dependencies, and can guarantee promising performances in many domains, e.g., natural language process (NLP) tasks [10, 11] and image tasks [12]. Similar for the domain of time series, it is also employed for wind power forecast [13]. Ref. [14] employed the network architectures in the transformer to conduct short-term wind power forecast considering spatio-temporal correlations; Ref. [15] uses a temporal transformer to facilitate multivariate and interpretable wind speed prediction, bringing substantial progress for wind speed prediction. In [16], a new transformer-based model is proposed for multivariate forecasting of wind speed and energy, showing superior performance than the baseline model. Except for these models, considering the characteristics of time-series signals, the signal decomposition technology [17], including empirical mode decomposition (EMD) [18], wavelet transform (WT) [19], Fourier transform [20], can be combined with deep learning-based models to extract more abundant feature from the raw data. For example, Ref. [21] combined EMD with GP-BP neural networks for wind speed forecast. Ref. [22] proposed EMD-based complex prediction model for wind power forecasting. Compared to only using deep learning-based models, the combination can obtain better results.

These studies mentioned generally optimize the accuracy of wind power forecasts by enhancing neural networks' ability to capture feature representations from wind power sequences. However, these studies primarily optimize the average prediction errors within a prediction horizon for focusing the overall performance of wind power, resulting in a loss of accuracy in the points within the LPI [23]. In addition, these studies present limited effectiveness in feature representation for the LPI, leading to poorer performance in those specific intervals. Accurately forecasting points within the LPI is crucial for controlling the specific reserve requirements of the power grid, thereby maximizing the utilization of wind power. Consequently, this paper proposes a advanced wind power forecast model to enhance the performance of wind power during the LPI.

Since the points within the LPI show closely correlated with ramp, including both the upward and downward ramps, this paper incorporates this association into the construction of the proposed model to gain a better understanding of the information related to the points within the LPI.

To sum up, our paper presents following contributions:

(1) this paper can enhance the accuracy of wind power prediction for the points within local peak interval by incorporating ramp slopes and magnitudes into the proposed model. To the best of our knowledge, this is among the first few studies to focus on the performance of local intervals.

(2) this paper introduces a multi-tasks prediction model that extends the traditional wind power forecasting model by incorporating a ramp identification task. This additional task supplies the traditional forecasting model with prior information regarding potential ramp events in the predicted wind power, thereby enhancing the accuracy of the points within local peak interval.

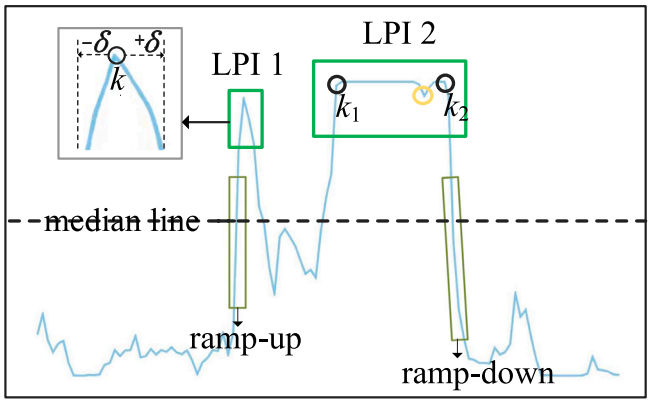


Fig. 1. The description of the LPI for wind power data.

(3) this paper employs learnable parameters to dynamically adjust the weights of each loss function, addressing the complex interdependencies among the multiple tasks of the proposed model. This approach enables a more flexible model, enhancing overall forecasting performance and offering an efficient solution to the multi-tasks nature of wind power prediction.

Our paper is presented: Section 2 describes problem statement; Section 3 describes preliminary, including problem definition; Section 4 describes the detailed information of the proposed model; Section 5 offers experiments on a wind power dataset. Finally, Section 6 makes conclusions.

2. Problem statement

For multi-step wind power forecasting, traditional models primarily maintain the similarity between the shape of actual and predicted wind power. These models prioritize overall accuracy across the entire multi-step prediction by assigning equal attention to each point, thereby overlooking the significance of specific intervals, ***e.g.***, ramp-down, ramp-up [24], and the points within the LPI. The description of these specific intervals are shown in Fig. 1. In power systems, these points exert substantial influences on the intraday dispatch of power system, and accurately forecasting them can ensure safety and economic operation of power grid. Due to the varying characteristics of wind power in these intervals, it is challenge to develop a generalized forecasting model that can improve the prediction accuracy for all key intervals. Instead, the proposed model primarily focuses on enhancing the accuracy of the points within the LPI. In this paper, we consider only two types of local peak intervals (LPI), such as LPI 1 and LPI 2 marked by green rectangle in Fig. 1. Within the type of LPI 2, there may occasionally be fluctuations, marked by yellow circle in Fig. 1. Due to the marginally small magnitude of these fluctuations, this study considers them to be part of the LPI 2.

Specifically for the LPI 1 and LPI 2, we present detailed mathematical descriptions. Considering a wind power data P, we calculate its median line by $\overline{P} = (P_{max} - P_{min})/2$. The definitions of the two LPI can be summarized in the following:

Type1 (**LPI1**): we firstly find the peak point k beyond the median line $\overline{P}$ by local maximum identification,

$$P(k) > P(t) + \epsilon \quad s.t., \quad t \in [k-\delta, k+\delta] \tag{1}$$

where δ is the range of the local maximum identification; ϵ is threshold; $P(t) > \overline{P}$. After find the peak point $P(k)$, we can define it together with its surrounding points with a size of δ $(P(k-\delta), \ldots, P(k-1), P(k+1), \ldots, P(k+\delta))$ to be LPI 1.

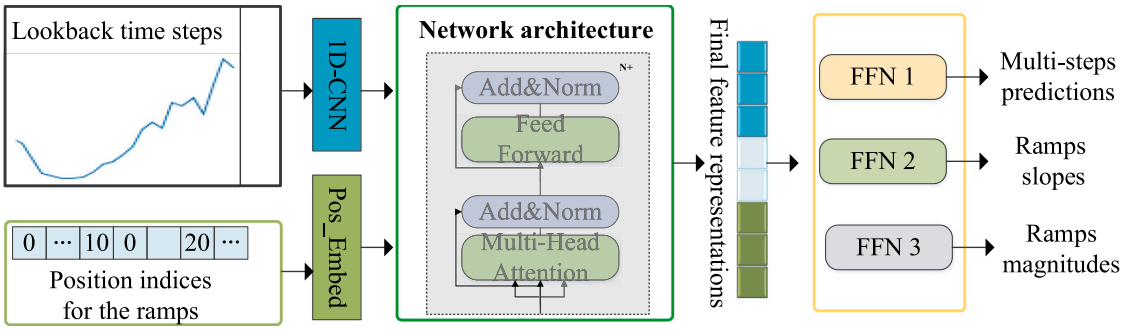


Fig. 2. The structure of the proposed model.

Type2 (**LPI2**): when a series of nearly parallel peaks is observed beyond the median line $\overline{P}$, we typically assign the label k_1 and k_2 to the peaks on the left and right side.

$$\begin{aligned} P(k_1) > P(t) + \epsilon \quad t \in [k_1 - \delta, k_1) \\ P(k_2) > P(t) + \epsilon \quad t \in (k_2, k_2 + \delta] \end{aligned} \tag{2}$$

where δ is the range of the local maximum identification; ϵ is threshold; $P(t) > \overline{P}$. Consequently, by applying these criteria, we can determine the points of interest in LPI 2 ($P(k_1 - \delta), \ldots, P(k_1), \ldots, P(k_2), \ldots, P(k_2 + \delta)$).

The difficulty in predicting the LPI arises from the substantial uncertainty and fluctuations associated with these points, making it challenging to accurately capture the information of them. Therefore, this paper proposes a novel model to enhance the capability of capturing information about the LPI, enabling to enhance the prediction accuracy of these points. Specifically, the points during the LPI is often influenced by the associated ramps, characterized by frequent changes in ramp slopes and magnitudes [25]. Additionally, the steepness of the ramp slopes and the size of the ramp magnitudes together determine the range of the LPI. The definition of the ramps is expressed by Eq. (3). Consequently, the proposed model incorporates a ramp identification task on top of the prediction task, thereby focusing its optimization on the points within the LPI.

$$|P(t + \Delta t) - P(t)| / \Delta t \geq R_{val} \tag{3}$$

where $P(t + \Delta t)$ and $P(t)$ are the values of wind power at the start and end of interval, respectively; R_{val} is a given threshold of ramp rate.

3. Preliminary

3.1. Problem formulation

In a typical multi-step WPF, the goal is to use a differentiable nonlinear function to predict future wind power based on data points within a lookback time span, denoted by $\boldsymbol{X} \in \mathbb{R}^{T \times 1}$, where T represents the length of the lookback window. This function can be formulated by $\mathcal{F}_\theta(\cdot)$ parameterized by θ. In addition to the $\boldsymbol{X}$, the input of the proposed model contains the position index of ramp occurrences within the $\boldsymbol{X}$, thereby producing another input $\boldsymbol{X}_{\mathrm{r}} \in \mathbb{R}^T$. In line with the multi-task approach of the proposed model, the output includes both regular multi-step predictions and detailed information about the ramps, including ramp slopes and magnitudes. The formulation of the proposed model can be expressed by,

$$\boldsymbol{Y}_{\mathrm{pre}}, \boldsymbol{Y}_{\mathrm{slope}}, \boldsymbol{Y}_{\mathrm{magnitude}} = \mathcal{F}_\theta(\boldsymbol{X}, \boldsymbol{X}_{\mathrm{r}}) \tag{4}$$

where $\boldsymbol{Y}_{\mathrm{pre}} \in \mathbb{R}^P$, $\boldsymbol{Y}_{\mathrm{slope}} \in \mathbb{R}^L$, and $\boldsymbol{Y}_{\mathrm{magnitude}} \in \mathbb{R}^L$ are the output of the predictions, ramp slopes, and ramp magnitudes, respectively.

4. Proposed algorithm

The proposed model is designed to enhance the performance of wind power especially for the LPI points. Since previously noted, the slope and magnitude of the ramp together determine the extent of the LPI points. Consequently, we incorporate the information of the ramp occurrences into the proposed model by encoding the position of them. Additionally, to enable the proposed model's ability to fit the LPI points, we add a slope recognition task into the standard prediction process, including the identification of the slope's gradient and magnitude. The structure of the proposed model is presented in Fig. 2.

4.1. Encoding: Integrating full time series with the position of ramps

Wind power forecasting is essentially a time series forecasting (TSF) problem, where data points in a time series are typically ordered. Incorporating positional information helps TSF models understand the position of each data point within the sequence, thereby improving their ability to interpret and predict future trends. Existing TSF models typically encode the position of all points within a time series. However, in this paper, we argue that encoding all points is not advantageous for predicting the LPI points in wind power data, particularly since predicting the LPI points is closely related to ramp information. Therefore, we focus on encoding only the points in the wind power sequence where ramps occur, based on Position Encoding [26]. In addition, wind power data presents periodicity due to the influence of meteorological factors like wind speed and direction. Consequently, how to capture the periodicity is significant in the prediction of wind power forecasting. Considering a fact that date features including year, month, day, and hour own natural periodic characteristics, we incorporate these features into the proposed model. Specifically, we combine the wind power and date features for each time point, thereby adding four additional dimensional features to the original data points $\boldsymbol{X}$. Subsequently, we map the combined features $\boldsymbol{X}_{\mathrm{c}} \in \mathbb{R}^{T \times 5}$ into a latent space using one-dimensional CNN [27] with a output dimension of d. The detailed calculations can be expressed by,

$$\boldsymbol{O}_{\mathrm{r}} = Position\ Encoding(\boldsymbol{X}_{\mathrm{r}}^t) = \begin{cases} \sin\left(\dfrac{\boldsymbol{X}_{\mathrm{r}}^t}{10000^{2j/d}}\right) \\ \cos\left(\dfrac{\boldsymbol{X}_{\mathrm{r}}^t}{10000^{2(j+1)/d}}\right) \end{cases} \tag{5}$$

$$\boldsymbol{O}_{\mathrm{c}} = 1DCNN(\boldsymbol{X}_{\mathrm{c}}^t) = \sum_{k=1}^{K} \boldsymbol{W}_k \cdot \boldsymbol{X}_{\mathrm{c}}^{t+k-1} + \boldsymbol{b} \tag{6}$$

where $\boldsymbol{X}_{\mathrm{c}}^t$ represents to the input of the combined wind power data $\boldsymbol{X}_{\mathrm{c}}$ at time t, where $t \in T$; Similarly, $\boldsymbol{X}_{\mathrm{r}}^t$ denotes the input of the original wind power ramp $\boldsymbol{X}_{\mathrm{r}}$ at time t, where $t \in T$; The parameter K corresponds to the kernel size of 1DCNN; $\boldsymbol{W} \in \mathbb{R}^{T \times d}$ and $\boldsymbol{b} \in \mathbb{R}^{1 \times d}$ is the weights and bias of 1DCNN, respectively; $j \in \{1, \ldots, \frac{d}{2}\}$, and d is the output channel of 1DCNN and Position Encoding; The output $\boldsymbol{O}_{\mathrm{c}} \in \mathbb{R}^{T \times d}$ corresponds to the original wind power data $\boldsymbol{X}_{\mathrm{c}}$ through

1DCNN, and the output $\boldsymbol{O}_{\mathrm{r}} \in \mathbb{R}^{T\times d}$ is the output from the positional encoding corresponding to the original wind power ramps $\boldsymbol{X}_r$; The final output $\mathbf{O}_{\mathrm{c,r}} \in \mathbb{R}^{T\times d}$ is obtained by combining $\boldsymbol{O}_{\mathrm{c}}$ and $\boldsymbol{O}_{\mathrm{r}}$ ($\boldsymbol{O}_{\mathrm{c,r}} = \boldsymbol{O}_{\mathrm{c}}+\boldsymbol{O}_{\mathrm{r}}$).

Subsequently, the output data $\boldsymbol{O}_{\mathrm{c,r}}$ will be fed into a deep learning network with advanced temporal modeling capabilities. This approach will facilitate the identification of latent patterns and complex features within the wind power data, thereby deepening our understanding of the underlying data structure and enhancing the model's prediction accuracy and robustness.

4.1.1. Network architectures

LSTM [28] and Transformer [29] are two prominent network architectures for sequential modeling, widely employed in various time series tasks. In this paper, we individually apply the two architectures to the proposed model and investigate their effects on the performance of the proposed model. This practice evaluates the generalization and adaptability of the proposed model.

LSTM networks are a specialized type of RNN designed primarily for modeling sequential data. The core of LSTM lies in its memory cells and gating mechanisms [30] (forget gate, input gate, and output gate), which collaboratively manage the selective retention and updating of information in time series data. A detailed mathematical formula can be expressed by,

$$\boldsymbol{F}_t = Sigmoid\left(\boldsymbol{W}_f \cdot [\boldsymbol{\gamma}_{t-1}, \boldsymbol{X}_t] + \boldsymbol{b}_f\right) \tag{7}$$

$$\boldsymbol{\sigma}_t = Sigmoid\left(\boldsymbol{W}_\sigma \cdot [\boldsymbol{\gamma}_{t-1}, \boldsymbol{X}_t] + \boldsymbol{b}_\sigma\right) \tag{8}$$

$$\tilde{\boldsymbol{C}}_t = TanhSigmoid\left(\boldsymbol{W}_c \cdot [\boldsymbol{\gamma}_{t-1}, \boldsymbol{X}_t] + \boldsymbol{b}_c\right) \tag{9}$$

$$\boldsymbol{C}_t = \boldsymbol{F}_t \cdot \boldsymbol{C}_{t-1} + \boldsymbol{\sigma}_t \cdot \tilde{\boldsymbol{C}}_t \tag{10}$$

$$\boldsymbol{O}_t = Sigmoid\left(\boldsymbol{W}_o \cdot [\boldsymbol{\gamma}_{t-1}, \boldsymbol{X}_t] + \boldsymbol{b}_o\right) \tag{11}$$

$$\boldsymbol{\gamma}_t = \boldsymbol{O}_t \cdot Tanh\left(\boldsymbol{C}_t\right) \tag{12}$$

where $\boldsymbol{F}_t$, $\boldsymbol{\sigma}_t$, and $\boldsymbol{O}_t$ are the output of the foreget gate, input gate, and output gate, respectively; $\boldsymbol{C}_t$ and $\boldsymbol{C}_{t-1}$ are the state of memory cells at the last and current time step, respectively; $\tilde{\boldsymbol{C}}_t$ is the candidate of memory cells; $\boldsymbol{\gamma}_{t-1}$ and $\boldsymbol{\gamma}_t$ are the output of state at the last and current time step; $\boldsymbol{W}_f$, $\boldsymbol{W}_\sigma$, $\boldsymbol{W}_c$, and $\boldsymbol{W}_o$ are the weights of these gate; $\boldsymbol{b}_f$, $\boldsymbol{b}_\sigma$, $\boldsymbol{b}_c$, and $\boldsymbol{b}_o$ are the bias of these gate; Sigmoid and Tanh are different activation function.

Traditional RNNs and LSTMs may encounter issues such as vanishing or exploding gradients when handling long sequences. In contrast, the transformer model utilizes an attention mechanism [31] to directly model relationships between any two positions in the sequence, effectively capturing long-range dependencies and mitigating gradient-related problems. Attention mechanism, defined by a tuple (query, key, value), focuses on global information in sequential data and presents a better performance in feature representations. We can calculate attention based on the tuple. The specific formulation can be expressed by,

$$Attention(\boldsymbol{X}_Q, \boldsymbol{X}_K, \boldsymbol{X}_V) = softmax(\frac{\boldsymbol{X}_Q \cdot \boldsymbol{X}_K}{\sqrt{d_k}}) \cdot \boldsymbol{X}_V \tag{13}$$

where $\boldsymbol{X}_Q$, $\boldsymbol{X}_K$, $\boldsymbol{X}_V$ are the element of the tuple, respectively; $softmax(\cdot)$ is activation function; d_k is the dimension of the element "key".

4.2. Identity ramps

After passing through a network architecture, we can obtain final feature representation of the original input $\boldsymbol{O} \in \mathbb{R}^{T\times d}$, where d is the output dimension of the network architecture. This process includes several layers of nonlinear transformations and feature extraction, enabling the network to extract significant features from the raw data. Consequently, this enhances the network's understanding concerning the input power, leading to more information for subsequent prediction and ramp recognition tasks for wind power.

Specifically, beyond a general forecasts of wind power over a specified prediction range, the proposed model incorporates two additional tasks: identifying the ramp slopes and magnitudes associated with the LPI points. In detail, followed by the obtain feature representations $\boldsymbol{O}$, we employ three distinct feed forward neural networks (FFN) [32] to respectively output general predictions, ramp slopes, and ramp magnitudes within a specific prediction range. The calculation can be expressed by,

$$\boldsymbol{Y} = ReLU(\mathbf{O} \cdot \boldsymbol{W}_{\mathrm{p}}) + \boldsymbol{b}_{\mathrm{p}} \tag{14}$$

$$\boldsymbol{Y}_{\mathrm{r,slope}} = ReLU(\mathbf{O} \cdot \boldsymbol{W}_{\mathrm{r,slope}} + \boldsymbol{b}_{\mathrm{r,slope}}) \tag{15}$$

$$\boldsymbol{Y}_{\mathrm{r,magnitude}} = ReLU(\mathbf{O} \cdot \boldsymbol{W}_{\mathrm{r,magnitude}} + \boldsymbol{b}_{\mathrm{r,magnitude}}) \tag{16}$$

where $\boldsymbol{Y}_{pre} \in \mathbb{R}^P$, $\boldsymbol{Y}_{\mathrm{r,slope}} \in \mathbb{R}^T$, and $\boldsymbol{W}_{\mathrm{r,magnitude}} \in \mathbb{R}^T$ are the output of regular predictions, ramp slopes, and ramp magnitudes, respectively. $\boldsymbol{W}_{\mathrm{p}} \in \mathbb{R}^{d\times P}$, $\boldsymbol{W}_{\mathrm{r,slope}} \in \mathbb{R}^{d\times L}$, and $\boldsymbol{W}_{\mathrm{r,magnitude}} \in \mathbb{R}^{d\times L}$ are the weights of these FFN; $\boldsymbol{b}_{\mathrm{p}} \in \mathbb{R}^{1\times P}$, $\boldsymbol{b}_{\mathrm{r,slope}} \in \mathbb{R}^{1\times L}$, and $\boldsymbol{b}_{\mathrm{r,magnitude}} \in \mathbb{R}^{1\times L}$ are the bias of these FFN; $ReLU$ is activation function.

4.3. Multi-task balanced loss function

In traditional forecasting tasks, mean squared error (MSE) loss function [33] is generally used to measure the errors between actual and predicted wind power. However, MSE is susceptible to noise and outliers, failing to capture the shape of time series data. To address the limitation, this paper employs SoftDTW [34], a loss function known for its ability to capture the shape of time series, to minimize the errors between the actual and predicted wind power. To ensure that the proposed model's prediction output come closer to the points within the LPI during training process, an additional task for ramp identification is incorporated . This task, includes determining both the magnitude and slope of the ramps, can guide the proposed model's optimization direction. Since both the slope and magnitude of the ramps are discrete, we employ the MSE to directly minimize the errors between the actual ramps and identified ramps. To address the complex inter-dependencies among these tasks, a dynamic weighted loss function based on learnable parameters is used to balance these tasks. More detailed information can be concluded in following contents.

4.3.1. SoftDTW loss function

The SoftDTW loss function defines an alignment matrix $\mathcal{A}_{T\times T} \subset \{0,1\}^{T\times T}$ to reflect the distance between the actual and predicted wind power points y and $\overline{y}$,

$$\mathcal{A}_{n,m} = \begin{cases} 1 & \text{if } y_n \text{is associated with } \overline{y_m} \\ 0 & otherwise, \end{cases} \quad n, m \in \mathbb{R}^T \tag{17}$$

Subsequently, due to the non-differentiability of the DTW algorithm, a cost matrix with a differentiable substitution-cost function is used to replace the DTW minimum, expressed by,

$$\Delta(y, \overline{y}) := \left[S(y_n, \overline{y_m})\right]_{nm} \in \mathbb{R}^{T\times T} \tag{18}$$

where S is a differentiable substitution-cost function; $\Delta(y, \overline{y})$ is a cost matrix.

Table 1
The statistical descriptions of wind power in these wind power datasets.

Statistical data	Numeric		
	Dataset 1	Dataset 2	Dataset 3
Mean (MW)	0.68	24.06	70.81
Standard Deviation (MW)	0.67	24.08	54.78
Percentile 25%	0.08	2.05	14.96
Percentile 50%	0.46	16.27	72.24
Percentile 75%	1.21	41.15	119.80
Number of LPI	1762	2019	1260

Table 2
The parameters used in the proposed model for each dataset.

Parameters	Numeric		
	Dataset 1	Dataset 2	Dataset 3
Train epoch	128	128	128
Batch size	256	256	256
Learning rate	0.0001	0.001	0.005
Size of surrounding points δ	2	2	2
Parameter τ in SoftDTW	2	4	6
Length of lookback time span T	12	24	24
Length of Predictions P	6	6	6
Length of Predictions L	6	6	6
Output dimension d	128	256	256

Table 3
The R2 of wind power predictions for LSTM and Transformer with and without the proposed model.

Datasets	LSTM	LSTM+	Transformer	Transformer+
Dataset 1	0.678	0.697	0.684	0.678
Dataset 2	0.688	0.692	0.685	0.691
Dataset 3	0.699	0.716	0.696	0.712

Table 4
The MAE and RMSE of the LPI points for LSTM and Transformer with and without the proposed model.

Dataset	Metric	LSTM	LSTM+	Transformer	Transformer+
Dataset 1	MAE	0.456	0.413	0.452	0.410
	RMSE	0.579	0.534	0.568	0.556
	MRE	1.449	1.420	1.420	1.330
Dataset 2	MAE	0.435	0.414	0.423	0.412
	RMSE	0.640	0.634	0.645	0.639
	MRE	1.070	1.065	0.916	0.906
Dataset 3	MAE	0.451	0.401	0.440	0.414
	RMSE	0.550	0.541	0.559	0.545
	MRE	1.054	1.028	1.056	0.965

Lastly, we conduct inner product on the $\mathcal{A}_{n\times m}$ and the $\Delta(y,\bar{y})$. Additionally, to make the SoftDTW loss differentiable, a generalized min operator with smooth parameter τ is used for this loss. The calculations of the SoftDTW loss can be expressed by,

$$\begin{aligned}\mathcal{L}_{softdtw} &= \mathrm{DTW}_{\tau}\left(y,\bar{y}\right) := \min_{\boldsymbol{A}\in\mathcal{A}_{n\times m}}\left\langle \boldsymbol{A}, \Delta\left(y,\bar{y}\right)\right\rangle \\ &= -\tau\log\left(\sum_{\boldsymbol{A}\in\mathcal{A}_{n\times m}}\exp\left(-\frac{\left\langle \boldsymbol{A}, \Delta\left(y,\bar{y}\right)\right\rangle}{\tau}\right)\right)\end{aligned} \tag{19}$$

where $\mathcal{A}_{n\times m}\subset\{0,1\}^{T\times T}$ is an alignment matrix; T is the length of a time series; $\boldsymbol{A}$ represents a path in $\mathcal{A}_{n\times m}$; y and $\bar{y}$ are the actual and predicted wind power, respectively.

4.3.2. MSE loss function

MSE loss function can optimize the average prediction errors between any two sequences y and $\bar{y}$. The calculation can be expressed by,

$$\mathcal{L}_{\mathrm{mse}} = \frac{1}{T}\sum_{n=1}^{T}(y_t-\bar{y}_t)^2 \tag{20}$$

where y_t and $\bar{y}_t$ are the actual and predicted wind power at time t, respectively;

4.3.3. Dynamic loss function

In response to the multi-tasks of the proposed model, we combine the SoftDTW loss function for the prediction of wind power and the MSE loss function for both the identification of the slope and magnitude of the ramps, thereby producing producing three optimization goals of $\mathcal{L}_{\mathrm{SoftDTW,pre}}(\mathcal{L}_1)$, $\mathcal{L}_{\mathrm{mse,slope}}(\mathcal{L}_2)$, and $\mathcal{L}_{\mathrm{mse,magnitude}}(\mathcal{L}_3)$, respectively. To balance them, a dynamic weighted loss function based on learnable parameters (σ_1, σ_2, and σ_3) is employed to adapt to the complex interconnection between these tasks,

$$\mathcal{L} = \frac{1}{2\sigma_1^2}\mathcal{L}_1 + \frac{1}{2\sigma_2^2}\mathcal{L}_2 + \frac{1}{2\sigma_3^2}\mathcal{L}_3 + \log\sigma_1\sigma_2\sigma_3 \tag{21}$$

where $\mathcal{L}$ is the final weighted loss function.

5. Experiment

5.1. Data description

To demonstrate the effectiveness of the proposed model, experiments were conducted on three wind power datasets.

(1) **Dataset 1**: this dataset is from a wind farms with 2 MW installed capacity in a region (longitude: −124 to −114; latitude: 32 to 42), available in U.S. Department of Energy's National Renewable Energy Laboratory (NREL) [35]. This is utilized by many researchers for wind power forecast-related studies. The dataset used for building the proposed model encompasses a span of one year, starting from January 1, 2013, to December 31, 2013, with a temporal resolution of 1 h.

(2) **Dataset 2**: this dataset is from a wind farm with 99 MW installed capacity in Western-North China, encompassing wind power data spanning a 2 year period, from January 1, 2019 to December 31, 2020. The temporal resolution of this dataset is 1 h.

(2) **Dataset 3**: the dataset originates from a wind farm located in North China with an installed capacity of 199 MW. It contains wind power data collected over a two-year period, spanning from January 1, 2019, to December 31, 2020, with a temporal resolution of 1 h.

To clearly demonstrate the differences among these three wind power datasets, the statistical information of them are calculated in Table 1, including mean, standard deviation, percentiles, and number of the LPI. It can be seen that wind power data presents significant fluctuations, presenting considerable challenges for traditional prediction models when dealing with intervals characterized by peaks in wind power. Consequently, it is essential to develop an advanced prediction model specifically designed to address the LPI.

To ensure sufficiently model training and retraining a portion of the data to evaluate the model's generalization ability, this paper divide it into a training and a testing sets by a date for each dataset, where the training dataset is used to train the proposed model and the testing model is used to measure the performance of the proposed model on wind power prediction. Specifically, the training dataset for Dataset 1, Dataset 2, and Dataset 3 are split by the follows: Dataset 1 covers from January 1, 2013, to October 1, 2013, while both Dataset 2 and Dataset 3 span from January 1, 2019, to October 1, 2020; For testing, Dataset 1 spans from October 2, 2013, to December 31, 2013, whereas Dataset 2 and Dataset 3 share the same testing period from October 2, 2019, to December 31, 2020.

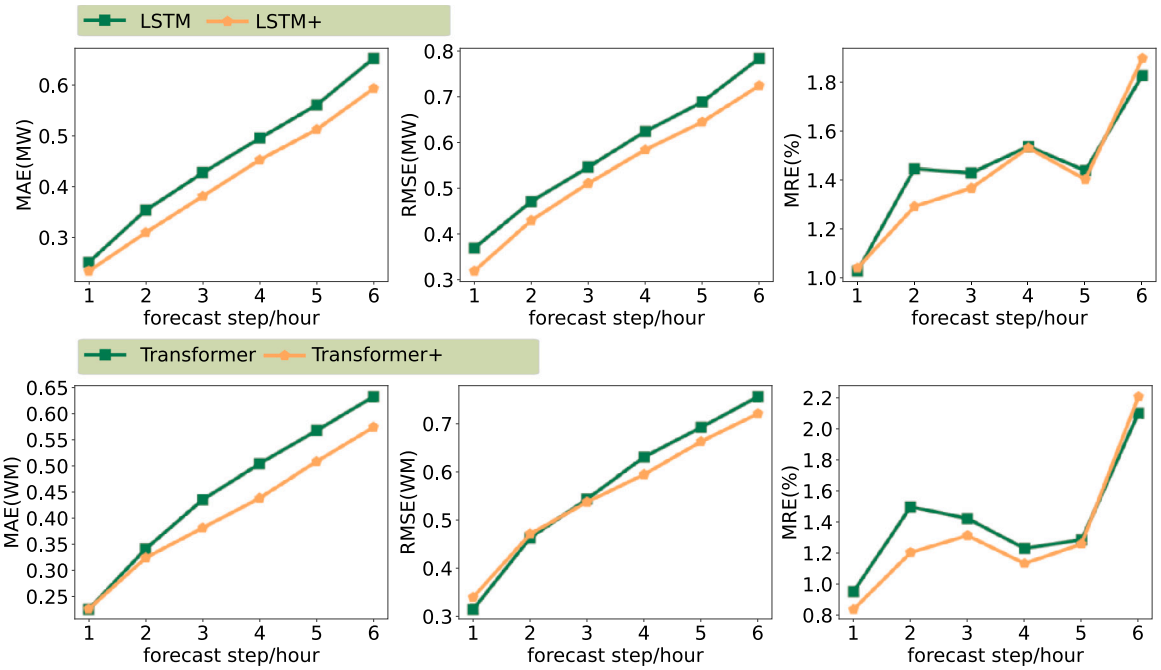


Fig. 3. The change of MAE, RMSE and MRE for both LSTM and Transformer with and without multi-tasks for Dataset 1.

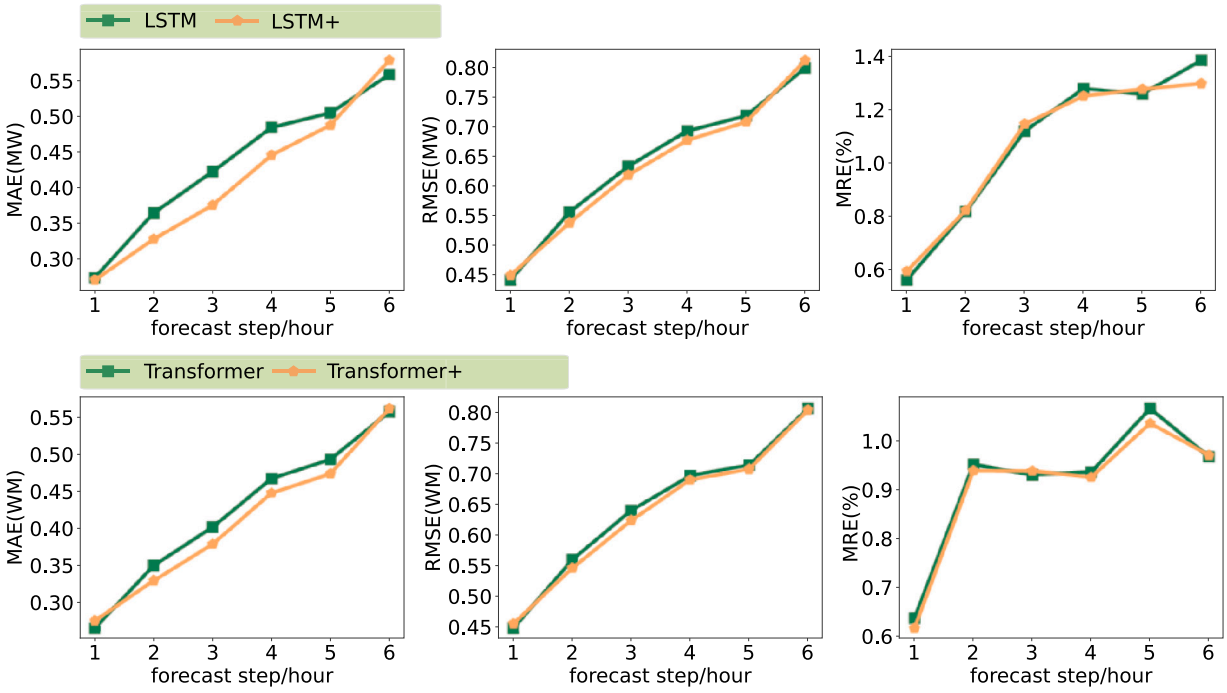


Fig. 4. The change of MAE, RMSE and MRE for both LSTM and Transformer with and without multi-tasks for Dataset 2.

5.2. Metrics and training settings

We make comparison analysis from two aspects: (1) one for the overall trend of the predicted and measured wind power using a metric of R-squared (R2); (2) another is for the points of LPI using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), and Mean Relative Error (MRE) due to the discrete nature of the points within the LPI.

All NNs were trained using Pytorch on a 3080 Ti GPU with 64 GB RAM, with optimized parameters learned by the backpropagation based Adam optimizer. The parameters of the proposed model used for each Dataset can be concluded in Table 2

5.3. Experimental analysis

Our paper demonstrates the effectiveness and robustness of the proposed model in forecasting wind power during the LPI through the following aspects: (1) the proposed model is a general method applicable to any end-to-end deep learning network architectures. We couple our model with multiple classic deep learning, including LSTM [30] and Transformer [29]; (2) we compare the proposed method with multiple

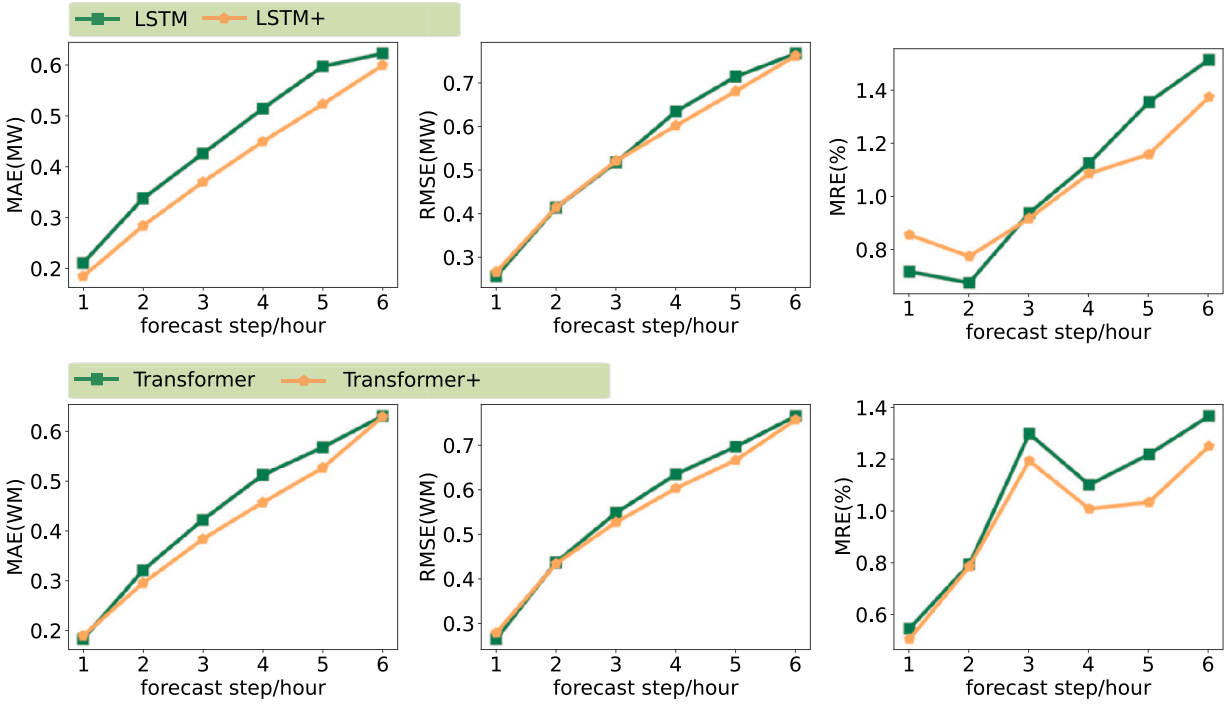


Fig. 5. The change of MAE, RMSE and MRE for both LSTM and Transformer with and without multi-tasks for Dataset 3.

Table 5
The network parameters of the SoTA models.

Model	Network parameters
RNN-GRU	RNN: hidden_size=256; num_layers=2 GRU: hidden_size=256; num_layers=2
Seq2Seq	Encoder: hidden_size=128; num_layers=2 Decoder: hidden_size=128; num_layers=2
Reformer	Encoder: d_model=256; n_heads=4, e_layers=2
LogTrans	sub_len=3; num_layers=2; n_embed=128
Informer	Encoder: d_model=512;n_heads=8;d_ff=512 Decoder: d_model=512;n_heads=8;d_ff=512

state-of-the-art backbone models in TSF, including RNN-GRU [36], Seq2Seq [37], Reformer [38], LogTrans [39] and Informer [40]. Additionally, we make parameter sensitivity study to demonstrate the importance of some parameters used in the proposed model on the performance of WPF, particularly for the points within the LPI.

5.3.1. Comparing NNs: With and without multi-tasks

The primary focus of our paper is not on optimizing network architectures to enhance performance at the LPI points. Instead, it is on facilitating ramp localization on the basis of predictions, enabling the proposed model to more accurately capture fluctuations in the LPI. This design can be flexible for any network architectures. Consequently, we apply the proposed model for multiple network architectures, including LSTM and Transformer, and conduct comparison analysis between these network architectures with and without the proposed model. We conduct two types of comparisons to measure the proposed model: (1) the R2 for the complete wind power predictions, and (2) the RMSE, MAE, and MRE for the LPI. The corresponding results of the three datasets are presented in Tables 3 and 4, respectively.

It can be seen from Tables 3 and 4 that a network architecture embedded with the proposed model not only ensures the overall trend of wind power but also enhance the prediction accuracy of the LPI points, compared to using the network architecture by itself. Specifically, in the prediction of the LPI points, the proposed model obtains significant results, with average enhancements of 7.10% in MAE, 2.66% in RMSE, and 3.46% in MRE, among the three Datasets. Specifically, the model enhances MAE by 9.35%, RMSE by 5.05%, and MRE by 4% for Dataset 1; for Dataset 2, the enhancements of MAE, RMSE, and MRE 3.64%, 0.9%, and 0.8%, respectively; Finally, Dataset 3 benefits from enhancements of 8.4% in MAE, 2.05% in RMSE, and 5.6% in MRE. Additionally, to present a clearer understanding of these enhancements over a prediction horizon of 6 h for each network architecture within the proposed model, the changes of these metrics are shown in Figs. 3, 4, and 5, corresponding to Dataset 1, Dataset 2, and Dataset 3, respectively.

5.3.2. Comparing SoTA models

We further compare the proposed model with the SoTA models, and analyze the gap of the performance of wind power forecast between them, including entire trend and the LPI. The network parameters are presented in Table 5.

In Fig. 6, we present the regression prediction curves of the proposed model and SoTA models on Dataset 1. The results indicate that both the proposed model and these SoTA models perform well in capturing the overall trend of wind power, accurately reflecting the original variations. However, upon further analysis, when we focus on the zoomed-in region of Fig. 6, particularly at the LPI points, it becomes clear that there are still significant discrepancies between the measured values and the predicted values of these SoTA models. These discrepancies reveal potential limitations in the SoTA models' ability to handle extreme values of wind power, especially at these points where prediction accuracy decreases. In contrast, the proposed model presents smaller errors at these LPI points and is able to better approximate the actual measured values. This suggests that the proposed model not only excels in capturing the overall trend but also significantly improves the accuracy of wind power predictions at extreme points compared to these SoTA models. This improvement may stem from the proposed model's more refined handling of extreme points or the inclusion of features better suited to capturing abrupt changes in wind power. Overall, this indicates that the proposed model offers higher

Table 6
The MAE and RMSE of wind power predictions for the proposed framework and SoTA models.

Datast	Metric	RNN-GRU	Seq2Seq	LogTrans	Reformer	Informer	Proposed
Dataset 1	RMSE (MW)	0.603	0.538	0.544	0.493	0.466	0.410
	MAE (MW)	0.720	0.671	0.694	0.629	0.650	0.555
Dataset 2	RMSE (MW)	0.445	0.448	0.474	0.470	0.447	0.412
	MAE (MW)	0.696	0.692	0.732	0.738	0.670	0.639
Dataset 3	RMSE (MW)	0.514	0.447	0.575	0.432	0.449	0.410
	MAE (MW)	0.624	0.541	0.721	0.549	0.575	0.546

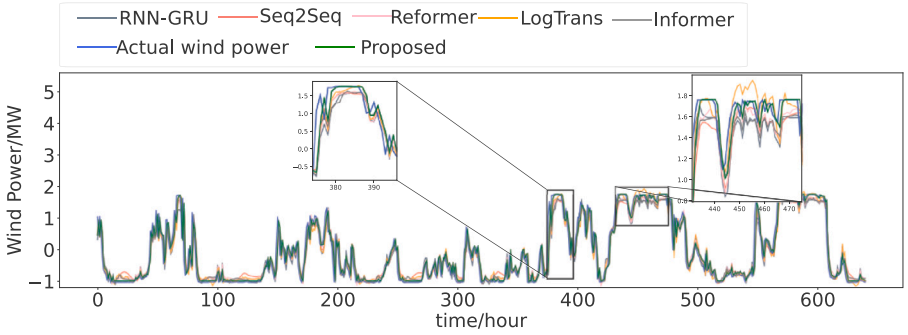


Fig. 6. The regression forecast curve of the proposed model and compared models.

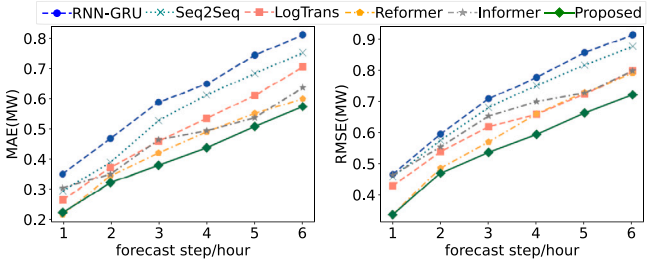


Fig. 7. The change of RMSE and MAE across forecast steps for Dataset 1.

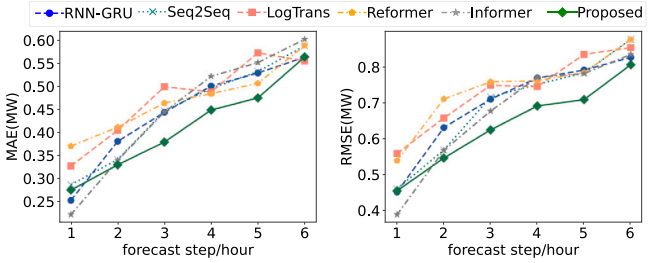


Fig. 8. The change of RMSE and MAE across forecast steps for Dataset 2.

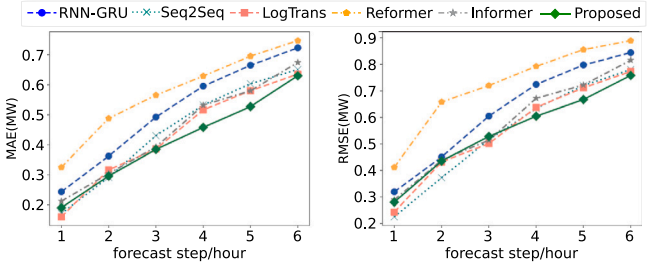


Fig. 9. The change of RMSE and MAE across forecast steps for Dataset 3.

robustness and reliability in wind power prediction, particularly when dealing with significant fluctuations in wind power.

In addition, similar for "Comparing NNs: with and without multi-task", we also compare the proposed model with these SoTA models from the two two aspects: (1) the R2 analysis for the trend of the complete wind power prediction, and (2) the RMSE, MAE, and MRE for the wind power during the LPI points. These metrics allow us to quantify the accuracy of the model's predictions at extreme wind power points and compare its performance with the SoTA models in addressing these critical points. Analyzing these aspects enables a more comprehensive evaluation of the proposed model's effectiveness across various prediction scenarios compared to current leading models.

Table 6 compares the average performance of our proposed model with the SoTA models at LPI points. Specifically, the proposed model shows a significant improvement in RMSE, with a performance gain of around 17.48% for Dataset 1, 9% for Dataset 2, and 7.09% for Dataset 3, compared to SoTA models, demonstrating its superiority in the prediction of wind power within the LPI points. For MAE, the proposed model also shows a notable enhancement, with a similar improvement of around 21.48% for Dataset 1, 9.20% for Dataset 2, and 12.60% for Dataset 3, further confirming its capabilities in reducing prediction errors of the LPI points. This result underscores the model's capability to tackle extreme wind power points, enhancing both prediction accuracy and the ability to manage extreme scenarios. Overall, these substantial enhancements presents the superior performance and reliability of the proposed model in wind power prediction tasks, offering more precise and robust results compared to current SoTA models.

To clearly shows the difference between the proposed model and the SoTA models, Figs. 7, 8, and 9 offer a detailed depiction of RMSE and MAE vary across different forecast steps for these Datasets. This comparison enables a direct observation of each model's performance trends and stability in different prediction steps. Specifically, with an increasing prediction step, the proposed model shows better forecasting accuracy compared to other models. Notably, at a longer forecast horizon of 6 h, the proposed model presents significantly lower error compared to current SoTA models, demonstrating superior accuracy and stability in managing extended time series data.

5.3.3. Parameter sensitivity analysis

We conducted an detailed analysis of the proposed method's parameters, specifically examining their influence on the model's performance

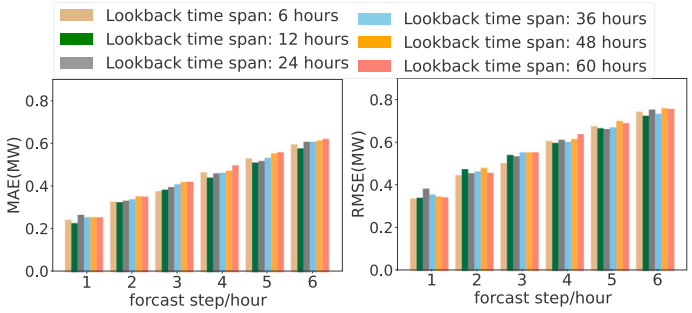


Fig. 10. The change of MAE and RMSE across different length of lookback time span for 1–6 h forecast steps.

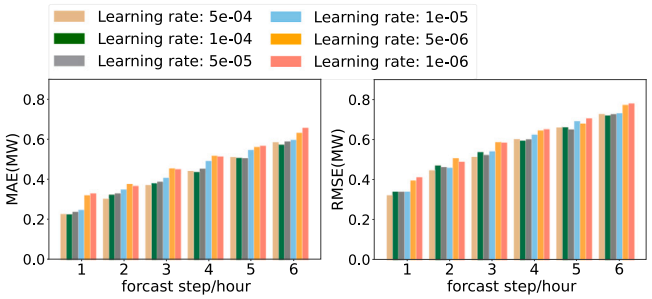


Fig. 11. The change of MAE and RMSE across different learning rates for 1–6 h forecast steps.

at LPI points. Our analysis focused on two key factors: the look-back time span and the learning rate. The look-back time span determines the amount of historical data the model can leverage, directly impacting its ability to capture trends of wind power. The learning rate influences both the convergence speed and the final optimization of the model. Careful selection and adjustment of these parameters are essential for maintaining the model's robustness and accuracy across various scenarios. By thoroughly analyzing these parameters, we can better understand the critical role of these parameters in enhancing model performance, thereby offering strong theoretical support for model optimization.

Lookback time span: we analyze the change of RMSE and MAE across lookback time span of 1–6 h forecast steps for wind power prediction at the LPI points, just seen in Fig. 10. With the length of lookback time span increasing, both RMSE and MAE gradually decrease, reflecting an improvement in the model's ability to capture historical data features. The model obtains optimal performance, with the prediction accuracy of 0.43 for MAE and 0.56 for RMSE at a lookback time span of 12 h. Nonetheless, when exceeding 60 h, performance begins to decline. This decline may be attributed to the inclusion of excessive redundant data, thereby increasing model complexity and weakens the model's ability to capture current data features.

Learning rate: we compared the performance of the model with different learning rates, including 5e−04, 1e−04, 5e−05, 1e−05, 5e−06, and 1e−06, and analyzed the impact of learning rates on MAE and RMSE for wind power prediction at the LPI points. Fig. 11 demonstrates that the proposed model obtains optimal prediction accuracy of 0.44 for MAE and 0.57 for RMSE when the learning rate is set below 0.0001. Nonetheless, once the learning rate exceeds 0.0001, the proposed model's accuracy becomes unstable and gradually declines. This finding presents that the learning rate presents a significant impact on model performance during specific time periods, requiring careful tuning in practical applications.

6. Conclusion

This paper proposes a novel wind power forecast model with dynamic ramp considerations to address the problem of worse performance of wind power during the LPI points. Specifically, on the basis of the function of the general forecast task, the proposed model performs an additional function to identity the ramps present in future predicted wind power. In addition, to manage the complex inter-dependencies among the multiple tasks, we employ learnable parameters to dynamically adjust the weights of the loss function in these tasks, thereby creating a more adaptable model for enhancing the performance of wind power within the LPI.

Specifically, our proposed model is flexible and can support various network architectures, e.g., LSTM, Transformer, and more. The RMSE, MAE, and MRE in the LPI can outperform the initial basis network architectures by a mean rate of 7.10%, 2.66%, and 3.46%, respectively, among three wind power datasets. In addition, when compared to SoTA models in predicting the points within the LPI, the proposed model demonstrates a remarkable improvement in RMSE, with a performance gain of approximately 17.48% for Dataset 1, 9% for Dataset 2, and 7.09% for Dataset 3. Similarly, for MAE, the proposed model presents significant enhancements, with improvements of around 21.48% for Dataset 1, 9.20% for Dataset 2, and 12.60% for Dataset 3. These findings demonstrate that the proposed model is powerful with accuracy and robustness in improving the performance of the LPI points for wind power.

Our paper opens up a new direction for wind power forecast that focus on the performance of the key local components in an entire wind power. This direction can be beneficial for the reserve requirements of power system, and ensure the safety and economic operation of power grid. Nonetheless, it just considers the LPI, and other key components, including unexpected up and down ramps, and sudden rise and drop fluctuations, can also be further studied in the future. Additionally, our paper all focus on a single wind power farm, and the power forecast of multiple wind plants should be further considered in the future.

CRediT authorship contribution statement

Nanyang Zhu: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Ying Wang:** Writing – review & editing, Writing – original draft. **Kun Yuan:** Writing – review & editing, Writing – original draft. **Jianhu Lv:** Writing – review & editing, Writing – original draft. **Bo Su:** Writing – review & editing, Writing – original draft. **Kaifeng Zhang:** Writing – original draft, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Our work was supported by State Grid Corporation of China Project (Research on Optimal Dispatch Considering Wide Bandwidth Power Regulation Under Multiple Operating Conditions After Three Changes of Coal power) (5108-202335059A-1-1-ZN).

Data availability

Data will be made available on request.

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